# Automatic Gain Control (AGC)

Complied By:
Kartik R. Patel
EXTC Department
KJSSE, SVU

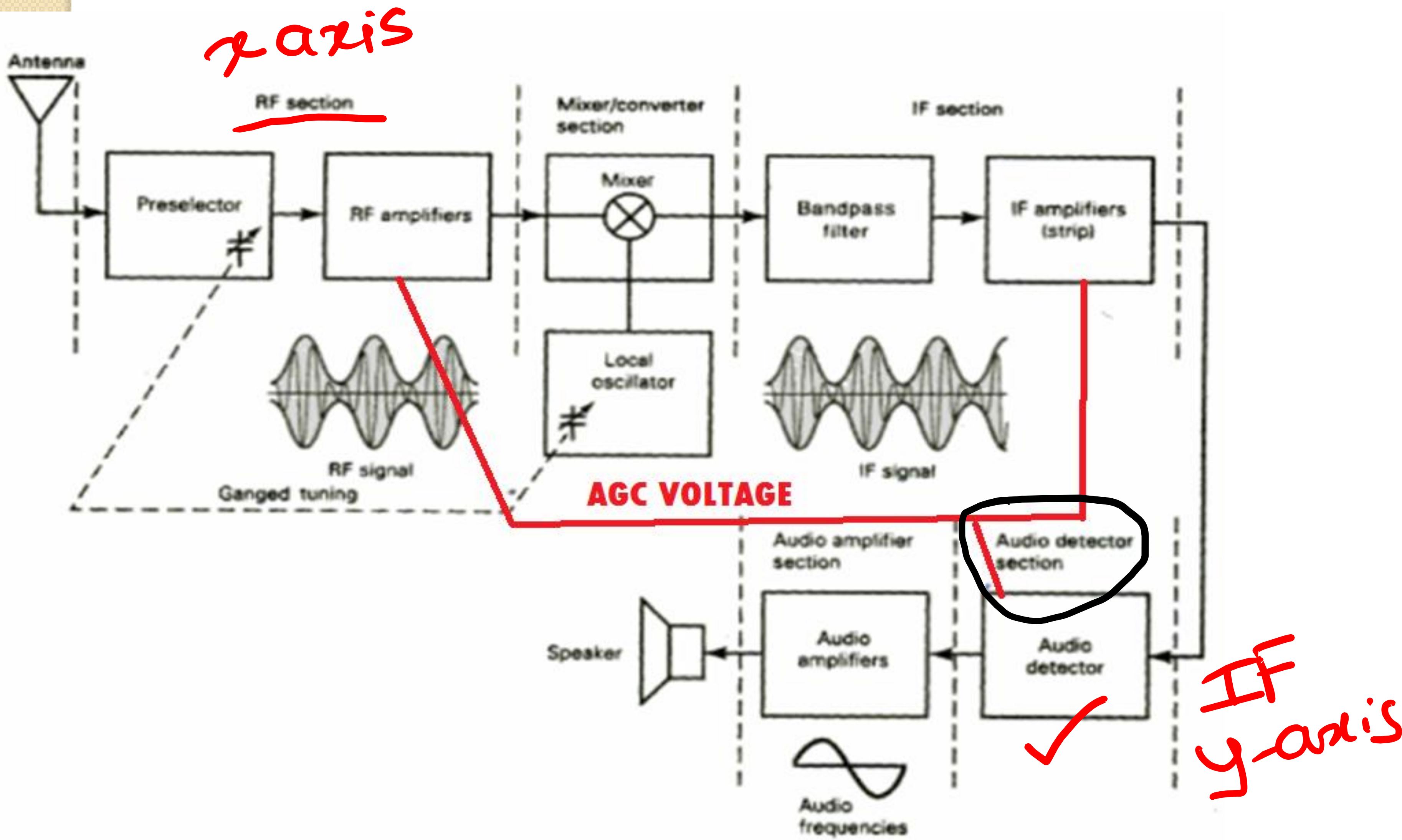
# Basic Principle

- An AGC circuit compensates for minor variations in the received RF signal level.

- It automatically increases receiver gain for weak RF input levels

- And automatically decrease the receiver gain when strong RF signal is received.

# Receiver with AGC



# Types of AGC

- Simple AGC
- Delayed AGC
- Forward AGC

# Simple AGC

- The AGC circuit monitors the received signal level and sends a signal back to the RF and IF amplifiers to adjust their gain automatically.

- AGC is a form of degenerative or negative feedback.

- The AGC circuit produces a voltage that adjusts the receiver gain and keeps the IF carrier power at the input to the detector constant.

# Delayed AGC

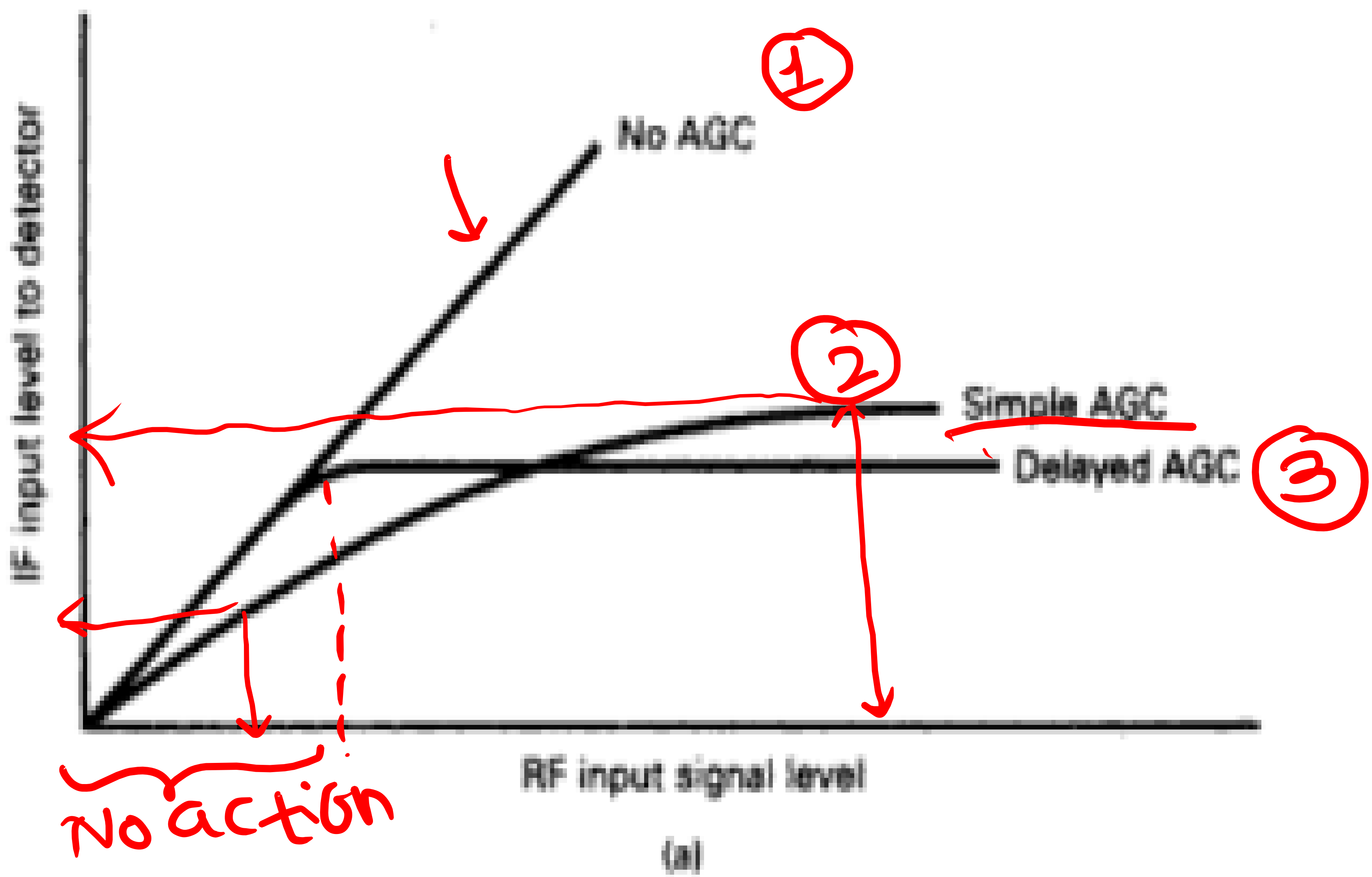
- With simple AGC, the AGC bias begins to increase as soon as the received signal level increases.

- Leads to false triggering, and receiver becomes less sensitive, which is called automatic desensing.

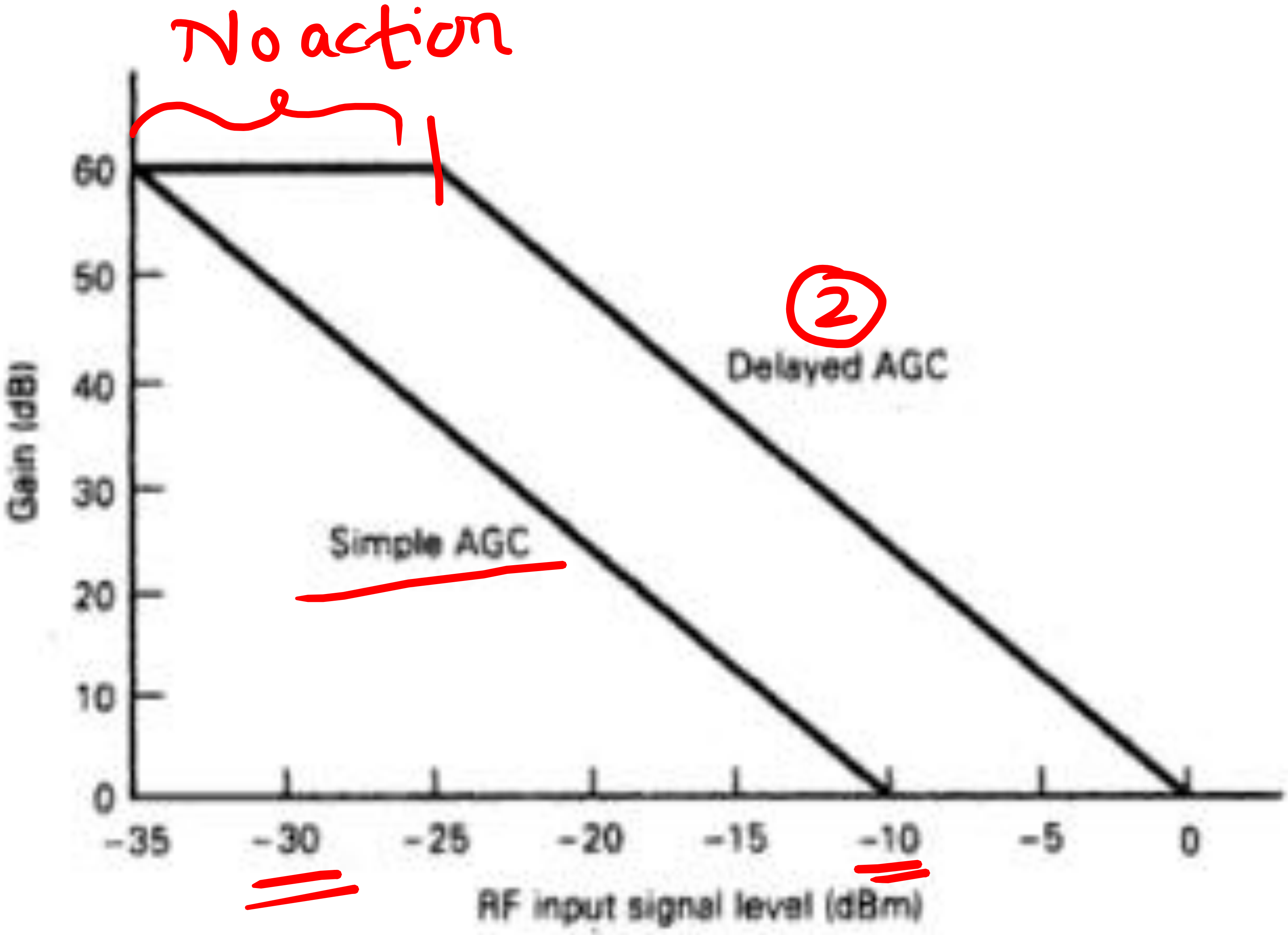
- Delayed AGC prevents feedback voltage until RF level exceeds a predetermined magnitude.

- After AGC threshold the output is proportional o carrier signal strength.

# AGC Response Characteristics



# AGC Response Characteristics



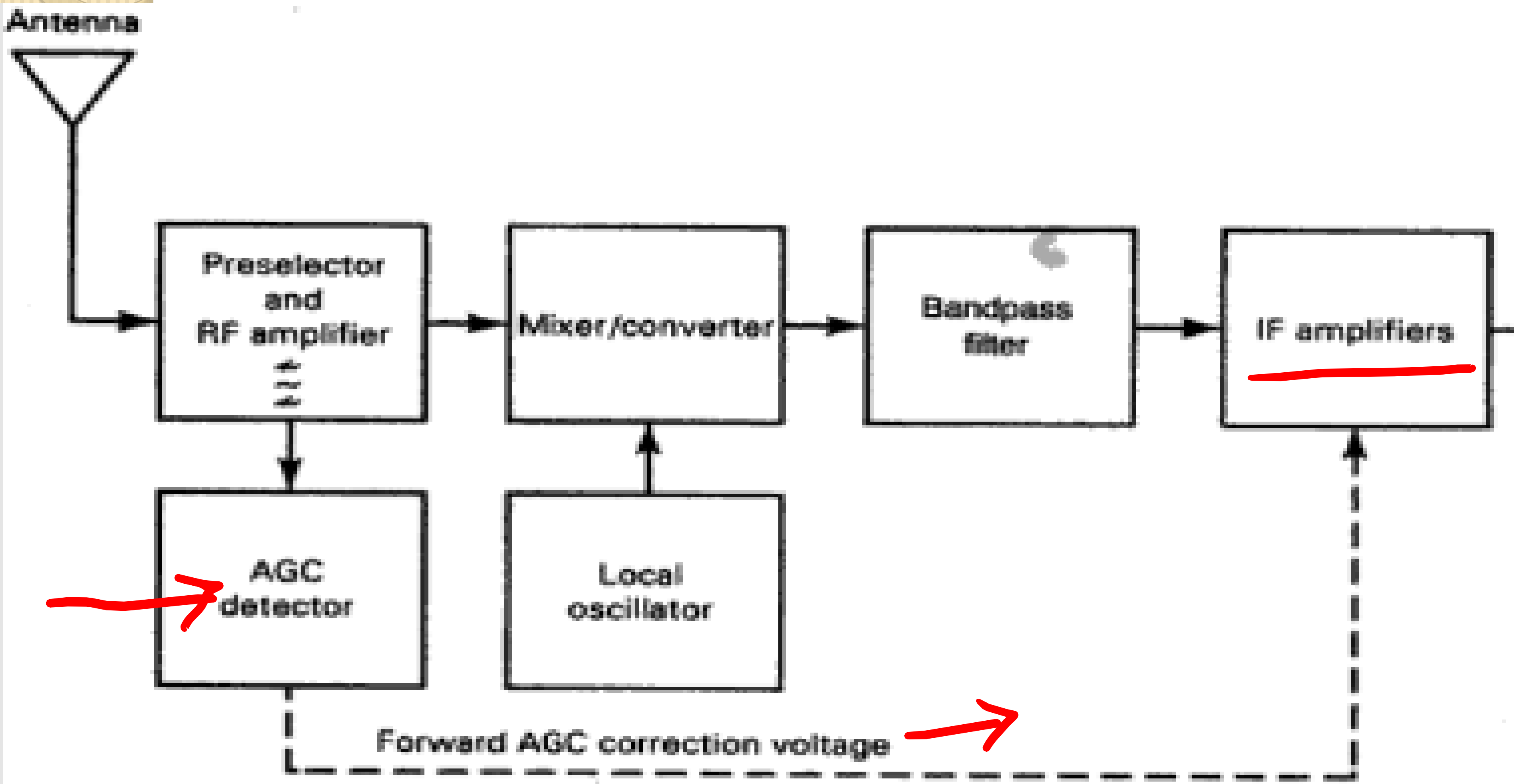
# Forward AGC

- Both delayed and simple AGC are post-AGC (after-the-fact) compensation.

- AGC voltage changes indicates that it may be too late.

- In forward AGC the received signal is monitored closer to the front end of the receiver and correction voltage is fed.

# Forward AGC


Antenna
Preselector and RF amplifier
Mixer/converter
Bandpass filter
IF amplifiers
AGC detector
Local oscillator
Forward AGC correction voltage